

Module 1

Numbers, Sets and Functions

1.1 Ordering and Inequalities

(1) Solve the following inequalities, with respect to x :

- (a) $\frac{2x}{x-1} \geq 6x-8$ (b) $4x^2 + 12x + 9 \geq x^2 - 2x + 1$; (c) $\sqrt{x^2+1} \leq \sqrt{2x+4}$;
(d) $x > \sqrt{4x-4}$; (e) $(2x-4)^2 \leq (x+3)^2$. (f) $\frac{x+1}{x-2} \leq \frac{x-1}{2x+3}$.

1.2 The induction principle

(1) Use the induction principle to verify that the following propositions are true

- (a) $2 + 4 + 6 + \dots + 2n = n(n+1)$ (b) $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
(c) $1 + 5 + 9 + \dots + (4n+1) = (2n+1)(n+1)$ (d*) Any set of n elements has 2^n subsets.

(2) Check that the following inequalities are true for $n = 1, 2$ and 3 . Then, use the induction principle to prove that they are true for every $n \in \mathbb{N}$.

- (a) $2^n \geq n+1$ (b) $3^n \geq n^2 - 1$ (c*) $3^n \geq n^3 - 1$

1.3 Domain and range of a function

(1) Find the natural domain D and the range R of the following functions. The natural domain is the largest possible set for which the function is defined.

(a) $y = \sqrt{x^2 - 1}$ (b) $y = x^2 + 3x - 1$; (c) $y = \ln(x^2 - 3)$;

(d) $y = \frac{\sin x}{1 - \cos x}$, for $-\pi \leq x \leq \pi$; (e) $y = 3^{1/x}$

(f) $y = \sqrt{(x-3)(x+2)(x-7)}$; (g) $y = \sqrt[4]{(x^2 - 4)}$;

(h) $y = \tan x - \frac{1}{x}$, for $-\pi \leq x \leq \pi$; (i) $y = \frac{x+1}{x-2}$;

1.4 Composition of functions

(1) Given the following functions $f(x)$ and $g(x)$, find the natural domain of the two possible compositions, i.e. $D_{f \circ g}$ and $D_{g \circ f}$:

(a) $f(x) = \ln x$, $g(x) = \sqrt{x+3}$ (b) $f(x) = 2^x$, $g(x) = x^2 + 3x + 1$

(c) $f(x) = \sqrt{-x}$, $g(x) = x^4 + x^2 + 6$ (d) $f(x) = \frac{x+2}{x-3}$, $g(x) = 3^x$

(2) Consider the following functions: $f(x) = \frac{1}{x}$, $g(x) = x^2 + 1$, $h(x) = 2x^2 - 3x + 4$, $w(x) = x^2$.

Write down expressions for the following compositions and determine the natural domain D of the resulting function:

(a) $h(h(x))$ (b) $f(\sin(w(x)))$ (c) $\ln(h(x))$
 (d) $f(\ln(h(x)))$ (e) $f(w(\sqrt{x} + 1))$ (f*) $f(g(h(x)))$

1.5 Inverse functions

(1) Find the inverses of the following functions, restricted to the indicated domains.

(a) $y = e^{(x-1)^2}$, for $x \geq 1$ (b) $y = \sqrt{x^3 - 1}$, for $x \geq 1$

(c) $y = \log_2(\ln x + 1)$, for $x > 1/e$ (d) $y = 2^{\sqrt{x}}$, for $x \geq 0$

(e*) $y = \frac{x^2 - 4}{x^2 + 1}$, for $x \geq 0$ (f) $y = x^2 - 3x + 2$, for $x \leq \frac{3}{2}$

(g) $y = -2x^2 + 2x + 1$, for $x \geq \frac{1}{2}$ (h) $y = \sqrt{x^2 + 1}$, for $x \leq 0$

1.6 Graphs of elementary functions

(1) Given that $|x|$ is the absolute value, $\text{sign}(x)$ is the sign, $\lfloor x \rfloor$ is the integer part and $\{x\}$ is the fractional part of x , sketch the graphs of the following expressions

$$(a) \quad y = \frac{x + |x|}{2} \quad (b) \quad y = |x - 2|; \quad (c) \quad y = |(x - 1)(x + 2)|;$$

$$(d) \quad y = \lfloor -x \rfloor \quad (e^*) \quad y = \lfloor x \rfloor - \lfloor -x \rfloor; \quad (f) \quad y = \text{sign}(x^2 - 4);$$

$$(g) \quad y = \{2x\} \quad (h) \quad y = 2\{x\}; \quad (i) \quad y = x \text{ sign}(x^2 - 1);$$

Part (e) is difficult but it might help to visualise it if you notice that the function is odd.

1.7 Composition of trigonometric functions with their inverses

(1) Find the exact values for the following expressions:

$$(a) \quad \arcsin\left(\frac{\sqrt{3}}{2}\right) \quad (b) \quad \arccos\left(-\frac{1}{2}\right) \quad (c) \quad \arctan(\sqrt{3})$$

$$(d) \quad \arccos\left(\frac{\sqrt{2}}{2}\right) \quad (e) \quad \sin\left(\arccos\left(\frac{\sqrt{3}}{2}\right)\right) \quad (f) \quad \sin\left(2 \arccos\left(-\frac{4}{5}\right)\right)$$

$$(g) \quad \cos\left(2 \arcsin\left(\frac{8}{17}\right)\right) \quad (h) \quad \tan\left(\arcsin\left(\frac{\sqrt{6}}{7}\right)\right) \quad (i^*) \quad \cos\left(\arctan\left(\frac{5}{3}\right)\right)$$

(Hint for question (i): Besides using the right-angled triangle sketch, the student can alternatively consider the expression for $\sin$ from the fundamental identity and divide both sides by $\cos x$ to find an expression of $\tan x$ in terms of $\cos x$.)

(2) Write the following as algebraic expressions in terms of x :

$$(a) \quad \sin(\arctan x) \quad (b) \quad \tan(\arccos x) \quad (c) \quad \cos(2 \arccos x)$$

$$(d) \quad \cos(\arctan x) \quad (e) \quad \sin(2 \arctan x) \quad (f) \quad \sec(\arctan x)$$

$$(g) \quad \text{cosec}(\arcsin x) \quad (h) \quad \cos(\text{arccosec } x) \quad (i) \quad \tan(\text{arcsec } x)$$

Take care in part (i). There is a subtlety connected with the sign.

1.8 Hyperbolic functions and their inverses

(1) Find the exact numerical values for the following expressions:

$$(a) \quad \sinh(\ln 4) \quad (b) \quad \sinh(-\ln 3) \quad (c) \quad \tanh(\ln \sqrt{3})$$

$$(d) \quad \text{sech}(\ln 2) \quad (e) \quad \text{cosech}(1) \quad (f) \quad \tanh^2(-1)$$

(2) Write the following as algebraic expressions in terms of x , simplifying when possible:

(a) $\tanh(2 \ln x)$ (b) $\sinh(\ln(1 - x))$ (c) $\cosh(\ln(1 + x))$

(3) If $\cosh x = \frac{5}{4}$, find the values of (a) $\sinh x$, (b) $\cosh 2x$ and (c) $\tanh x$.

(4) If $\sinh x = -\frac{2}{5}$, find the values of (a) $\cosh x$, (b) $\tanh x$ and (c) $\operatorname{sech} x$.

(5)

(a) $\cos^2(x) + \sin^2(x) = 1$

(b) $\cos^2(x) - \sin^2(x) = \cos(2x)$

(c) $\sin(2x) = 2 \sin(x) \cos(x)$

(d) $1 + \tan^2(x) = \sec^2(x)$

(e) $1 + \cot^2(x) = \operatorname{cosec}^2(x)$

(f) $\sin(x + y) = \sin(x) \cos(y) + \cos(x) \sin(y)$

(g) $\cos(x - y) = \cos(x) \cos(y) + \sin(x) \sin(y)$

(h) $\tan(x + y) = \frac{\tan(x) + \tan(y)}{1 - \tan(x) \tan(y)}$

(i) $\cos(3x) = 4 \cos^3(x) - 3 \cos(x)$

(j) $\sin(3x) = 3 \sin(x) - 4 \sin^3(x)$

(k) $\sin(4x) = 4 \cos(x) (\sin(x) - 2 \sin^3(x))$

(l) $\sin(5x) = 5 \sin(x) - 20 \sin^3(x) + 16 \sin^5(x)$

(6) Use the definitions of $\cosh x$ and $\sinh x$ to verify the following identities:

(a) $\cosh(2x) - \sinh(2x) = e^{-2x}$

(b) $\cosh^2 x + \sinh^2 x = \cosh(2x)$

(c) $(\cosh x + \sinh x)^{100} = \cosh(100x) + \sinh(100x)$

(d*) $\tanh x - \frac{1}{\tanh x} = -2 \operatorname{cosech}(2x)$

(7a) On the same plot, sketch the graphs of the following functions:

$$y = \cosh x, \quad y = \sinh(x - 3), \quad y = \operatorname{cosech}(x - 3).$$

(7b) On the same plot, sketch the graphs of the following functions:

$$y = \sinh 3x, \quad y = \cosh(3x + 2), \quad y = \operatorname{sech}(3x + 2)$$

(8) (*Multiple to powers*) Using the definition of $\sinh x$ and $\cosh x$, prove the following relations:

(a) $\sinh(4x) = (4 \sinh x + 8 \sinh^3 x) \cosh x$

(b) $\cosh(3x) = 4 \cosh^3 x - 3 \cosh x$

(9) (*Powers to multiple*) Using the definition of $\sinh x$ and $\cosh x$, prove the following relations:

$$\begin{aligned} \text{(a)} \quad \sinh^4 x &= \frac{\cosh 4x - 4 \cosh 2x + 3}{8} \\ \text{(b)} \quad \cosh^3 x &= \frac{\cosh 3x + 3 \cosh x}{4} \end{aligned}$$

(10) Using the definitions of $\operatorname{arcsinh} x$, $\operatorname{arccosh} x$ and $\operatorname{arctanh} x$, rewrite the following as algebraic expressions in x :

$$\begin{aligned} \text{(a)} \quad & \cosh(\operatorname{arcsinh} x), & \text{(b)} \quad & \sinh(2 \operatorname{arctanh} x), \\ \text{(c)} \quad & \operatorname{sech}(3 \operatorname{arccosh} 2x), & \text{(d)} \quad & \cosh(2 \operatorname{arctanh} x). \end{aligned}$$

(11) On the same plot, sketch the graphs of the following functions:

$$y = \cosh\left(\frac{x}{2}\right), \quad y = \operatorname{sech}\left(\frac{x}{2}\right), \quad y = \operatorname{arccosh}\left(\frac{x}{2}\right).$$

(12) On the same plot, sketch the graphs of the following functions:

$$y = \sinh(x-1), \quad y = \operatorname{cosech}(x-1), \quad y = \operatorname{arcsinh}(x-1).$$